Meme Coin Analysis Dashboard

Data Engineering Project Proposal



much data.
such pipeline.

IDSC501

Team members: TBD

Doge illustration: Icons8

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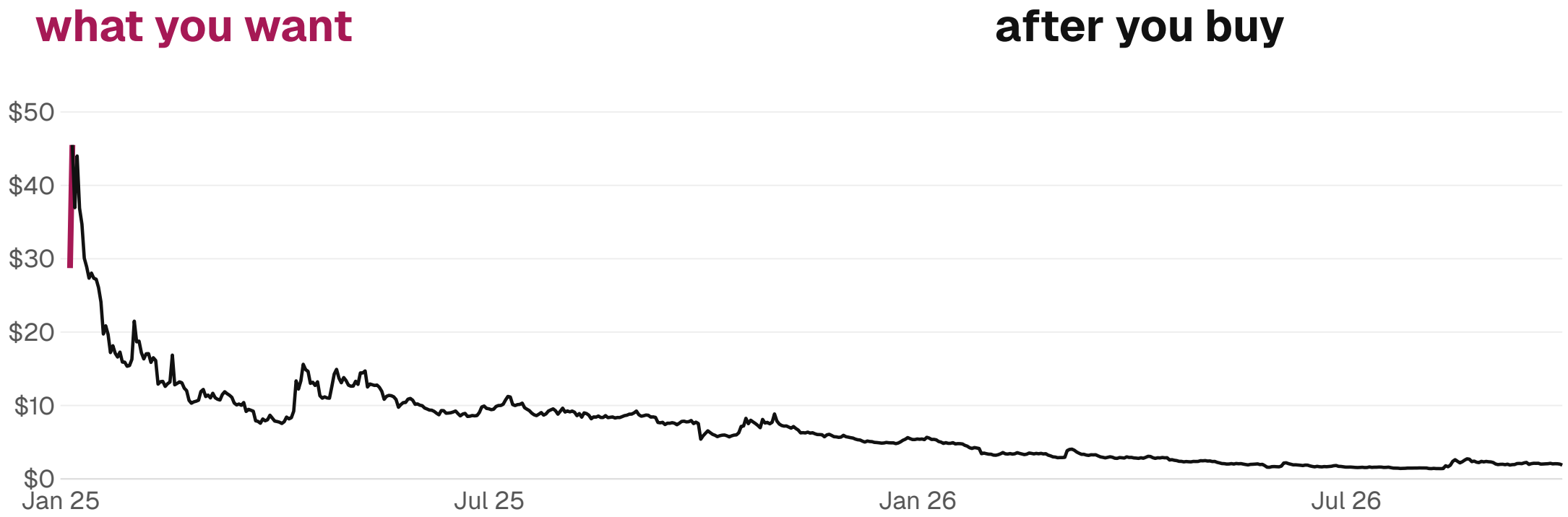
02 **Data collection** — Sources and access methods

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04 **Database & application** — Storage, joins and the dashboard

Why meme coins?

TRUMP daily close (USD)



What was the community saying?

18 Jan 2025–7 Oct 2026. Supplied daily closes. Captions are a joke, not an actual trade.

Data overview

Market data

Structured records

14,773

daily records across 10 coins

OHLCV and market cap
UTC date

Social data

Text + platform metadata

Posts + comments

Text and creation time
Observed reactions
Account or community

Prices available

Social collection planned

Price input ends on 7 Oct 2026. Selected X accounts and eligible Reddit communities define the social scope.

Data collection

	Source	Collection method	Output
	CoinMarketCap Historical Data	**Web crawling**	Daily price CSV
	X via FxTwitter	**Third-party API**	Posts + reactions in JSON
	Reddit via Arctic Shift	**Archive API / files**	Posts + comments

Daily social batches, with a seven-day overlap.

Project websites: crawl links and contract details to verify sources.

Reddit awaits access conditions. Price refresh needs new CSVs or restoration of the team crawler.

Data preprocessing

From raw posts to daily features.

Raw JSON

```
{
  "id": "<post_id>",
  "text": "$DOGE…",
  "created_at": "<UTC>",
  "likes": null
}
```

Standardize

Stable post ID
UTC timestamp
Language + repost flag
Verified coin match

Daily features

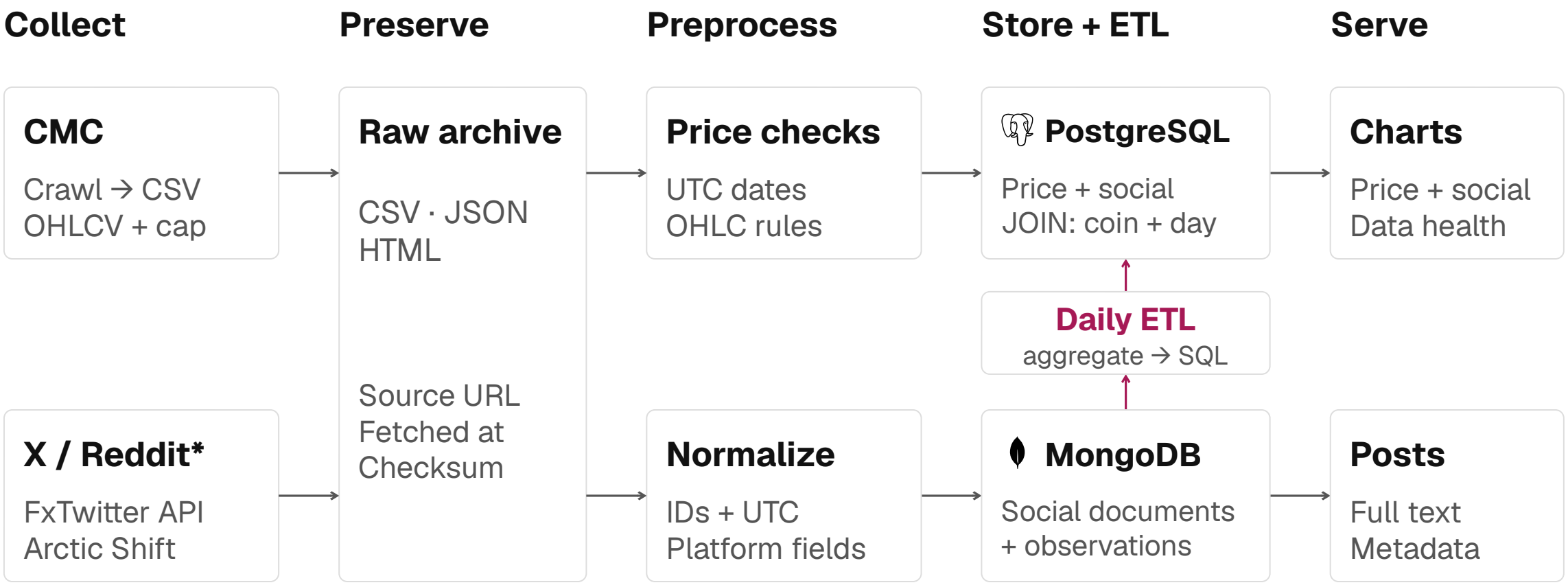
Coin + UTC date
Observed post count
Mean sentiment
Coverage status

Price checks Duplicate days, OHLC rules, missing or zero values

NULL ≠ 0 Collection failure stays missing.

Schema illustration. Sentiment uses supported English text. Preserve raw inputs for replay.

Data pipeline & storage



* Reddit: conditional access. Social: daily batch; 7-day replay. Archive: monthly.

Streamlit + Plotly

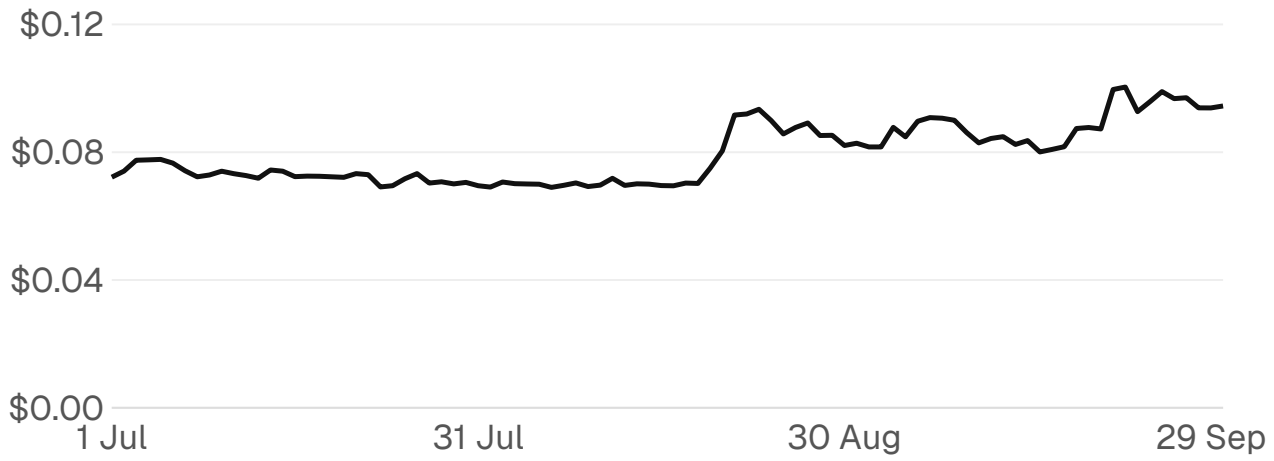
MongoDB documents → ETL → SQL analytics

PostgreSQL also supports JSON. Two stores make the models explicit and add operating costs.

Dashboard & expected outcome

Collect → Prepare → Store → Explore

DOGE 2026 Q3 Concept preview



Social activity

Awaiting eligible collection

Coverage: unavailable

Compare time series

Inspect posts

Check coverage

Streamlit + Plotly. Charts query PostgreSQL. Post details query MongoDB. DOGE prices use supplied data.